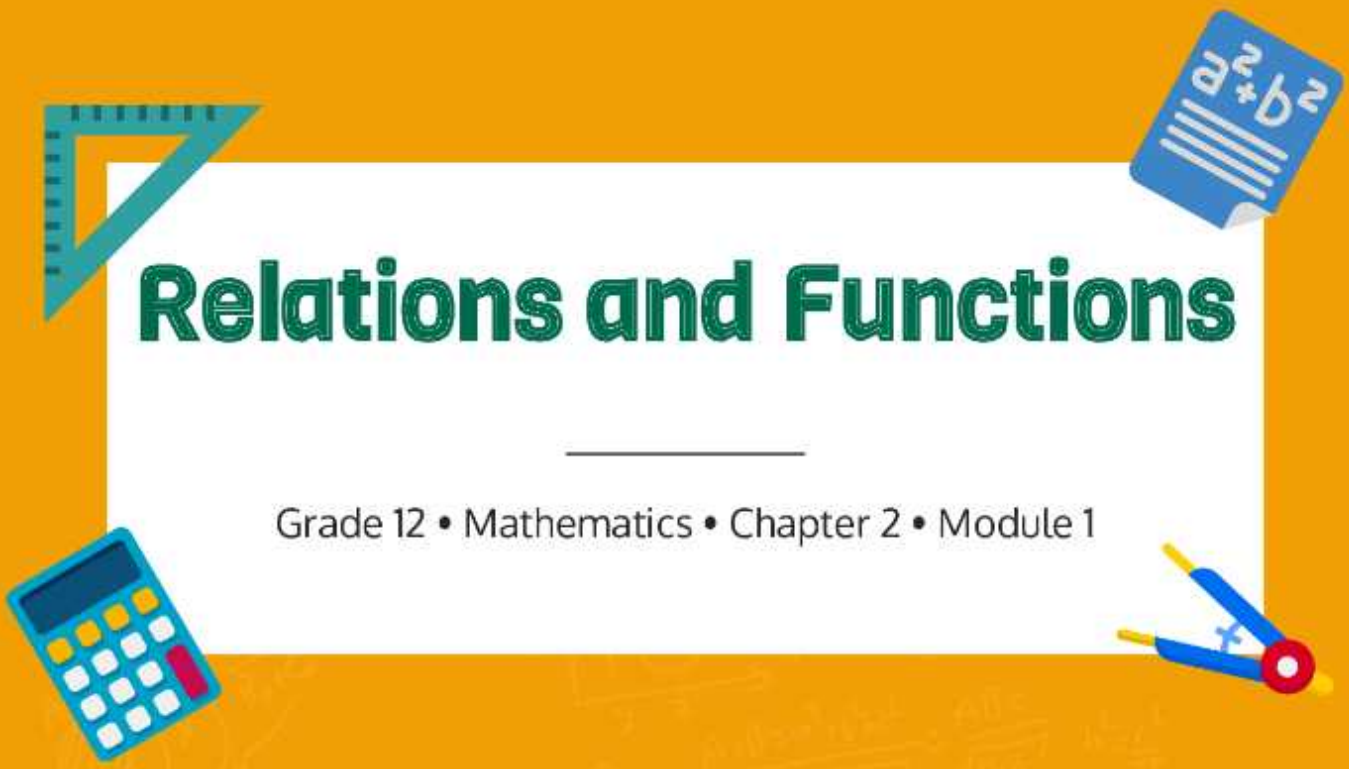

Relations and Functions
Grade 12 • Mathematics • Chapter 2 • Module 1



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1.2 Relations







1.2 Relations

There are three ways of writing a relation

Let us do it by examples

Let A = {1, 2, 3} and B = {1, 3, 5, 7}

(1) R = {(1,1), (2, 3), (3, 5)}

(or)

(2) R = $\{(a, b) \mid b = 2a - 1, a \in A \text{ and } b \in B\}$

(or)

(3) a R $b \Leftrightarrow b = 2a - 1$; $a \in$ A and $b \in$ B

1.3 Number of Relations

Result

If $n(A) = p$ and $n(B) = q$ then the number of relations that can be defined from A to B is 2^{pq}.

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1.4 Inverse of Relations

If R is a relation on set A, then the relation R^{-1} on A ,defined by

$R^{-1} = \{(b, a) : (a, b) \in R\}$ is called the inverse relation on A.

Clearly,

Domain (R^{-1}) = Range of (R)

Range of (R^{-1}) = domain of (R).

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$A = \{1, 2, 3\}$

$B = \{5, 7, 9\}$

$R : A \to B$ $\quad (a, b)$

$R = \{(1,5), (2,7)\}$

$R^{-1} : B \to A$ $\quad (b, a)$

$R^{-1} = \{(5,1), (7,2)\}$

1.4 Inverse of Relations

If R is a relation on set A, then the relation R^{-1} on A ,defined by

$R^{-1} = \{(b, a) : (a, b) \in R\}$ is called the inverse relation of A.

Clearly,

Domain (R^{-1}) = Range of (R)

Range of (R^{-1}) = domain of (R)

Ex $R = \{(a, 1), (b, 2), (c, 3), (d, 4)\}$

$R^{-1} = \{(1, a), (2, b), (3, c), (4, d)\}$

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TYPES OF RELATION

i) IDENTITY RELATION (I_A)

$R : A \to A$

$A = \{1, 2, 3\}$

$R = \{(1,1), (2,2), (3,3)\}$

$R_1 = \{(1,1), (2,2), (3,3), (1,3)\}$

$R_2 = \{(1,1), R_{12}\}$

No

X

ii) REFLEXIVE RELATION

$R : A \to A$

$A = \{1, 2, 3\}$

$R_1 = \{(1,1), (2,2), (3,3), (1,3)\}$ ⟵ (R)

$R_2 = \{(1,1), (2,2), (1,3)\}$ ✗

$R_3 = \{(1,1), (2,2), (3,3)\}$ ⟵ (R)

2 Types of Relations

(3) Identity Relation

If every element of A is related to itself only. Symbolically, the relation

$I_A = \{(a, a) : a \in A\}$ is called an Identity relation on A.

For example, let A = {1, 2, 3} then relation R_1 on A , defined as $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ is an identity relation.

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2 Types of Relations

(3) Identity Relation

If every element of A is related to itself only. Symbolically, the relation

$I_A = \{(a, a) : a \in A\}$ is called an Identity relation on A.

For example, let A = {1, 2, 3} then relation R_1 on A , defined as $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ is an identity relation.

Note

$R_2 = \{(1, 1), (2, 2)\}$ and $R_3 = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$ are not identity relations on A.

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2 Types of Relations

(4) Reflexive Relation

A Relation R on a set A is said to be reflexive, if every element of A is related to itself.
Thus if R is reflexive, then $(a, a) \in R, \forall\ a \in A$.

For example, let A = {1, 2, 3} then $R_1 = \{(x, y) \mid y \geq x\}$ is a reflexive relation on A.

$x=1;\ y \geq 1 \quad \{x=2 \quad y \geq 2\} \quad x=3 \quad y \geq 3$

$R_1 = \{(1,1), (1,2), (1,3); (2,2), (2,3), (3,3)\}$

SYMMETRIC RELATION

$R : A \to A$

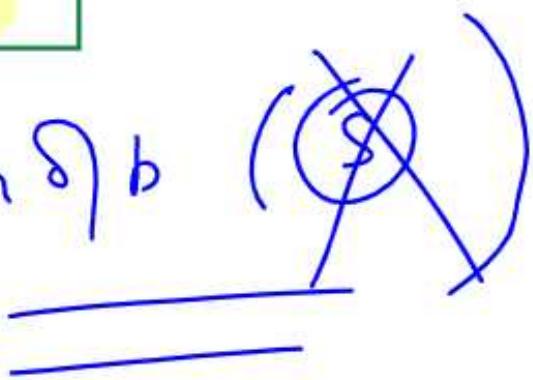
$A = \{1, 2, 3\}$ $\quad (a, b) \Rightarrow (b, a)$

$R_1 = \{(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2)\}$

$L_1 \parallel^{Rel} L_2 \Rightarrow L_2 \parallel^{Rel} L_1 = (S)$

$L_1 \perp^R L_2 \Rightarrow L_2 \perp^R L_1$

$aRb \Rightarrow$ a & b brother



2 Types of Relations

(5) Symmetric Relation

A relation R on a set A is said to be a symmetric relation iff
$(a, b) \in R \Rightarrow (b, a) \in R, \forall\ a, b \in A$
i.e. $a\ R\ b \Rightarrow b\ R\ a, \forall a, b \in A$

For example, let A = {1, 2, 3} then R_1 = {(1, 2), (1, 3), (2, 2), (2, 1), (3, 1)} is a symmetric relation on A.

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2 Types of Relations

(5) Symmetric Relation

A relation R on a set A is said to be a symmetric relation iff
$(a, b) \in R \Rightarrow (b, a) \in R, \forall\ a, b \in A$
i.e. $a\ R\ b \Rightarrow b\ R\ a, \forall a, b \in A$

Ex Let A = {1, 2, 3} then R_1 = {(1, 2), (1, 3), (2, 2), (2, 1), (3, 1)} is a symmetric relation on A.

Ex Let A be the set of all lines in a plane. Relation R defined on A as $L_1\ R\ L_2 \Leftrightarrow L_1 \perp L_2$ is a symmetric relation.

Let's Solve

Ex. Prove that the relation defined as

$R = \{(a, b) : a - b \text{ is divisible by } 4,\ a, b \in \mathbb{Z}\}$ is symmetric.

$(a, b) \Rightarrow a - b = 4m \quad m \in I$

$(b, a) \Rightarrow b - a = -4m$

$b - a$ is divisible by 4

S

Let's Solve

Ex. Prove that the relation defined as

$R = \{(a, b) : a - b \text{ is divisible by } 4,\ a, b \in \mathbb{Z}\}$ is symmetric.

Solution

Let $(a, b) \subset R$

$\Rightarrow a - b = 4k,$ for some $k \in \mathbb{Z}$

$\Rightarrow b - a = 4(-k)$

$\Rightarrow b - a$ is also divisible by 4

$\Rightarrow (b, a) \in R$

$\therefore$ R is symmetric

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Let's Solve

(3,3), (4,4) (4,3), (3,2)

Ex. Let A = {1, 2, 3, 4} and R = {(1, 1), (2, 2), (3, 4), (2, 3)}. At least how many elements need to be added in R so that it becomes reflexive and symmetric?

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Let's Solve

Ex. Let A = {1, 2, 3, 4} and R = {(1, 1), (2, 2), (3, 4), (2, 3)}. At least how many elements need to be added in R so that it becomes reflexive and symmetric?

Solution

Let A = {1, 2, 3, 4} and R = {(1, 1), (2, 2), (3, 4), (2, 3)}

To make it reflexive,

We have to add (3, 3), (4, 4)

And to make it symmetric

(4, 3), (3, 2)

So in total 4 elements need to be added.

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TRANSITIVE RELATION:

$R : A \to A$

$(x, y), (y, z) \Rightarrow (x, z)$

2 Types of Relations

(6) Transitive Relation

Relation R on set A is transitive if xRy and $yRz \Rightarrow xRz$

Remark

Relation is not transitive only when xRy and yRz but not xRz. In all other situations R is Transitive.

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2 Types of Relations

(6) Transitive Relation

Relation R on set A is transitive if xRy and $yRz \Rightarrow xRz$

Let A = {1, 2, 3, 4, 5}

(1) R_1 = {(1, 2), (2, 3), (1, 3)} It is transitive relation.

(2) R_2 = {(1, 2), (2, 3), (1, 3), (4, 5), (5, 1)} It is not a transitive relation.

(3) R_3 = {(1, 3), (2, 4)} is transitive.

Let's Solve



Let's Solve

Ex. Prove that relation R = $\{(x, y) : x \text{ is factor of } y\}$ is a transitive relation on integers.

$y = \lambda_1 x \quad \lambda_1 \in I$

$(x, y)\ ;\ (y, z) \Rightarrow (x, z)$

$y = \lambda_1 x$

$z = \lambda_2 y$

$\Rightarrow z = \lambda_2(\lambda_1 x)$

$z = (\lambda_1 \lambda_2) x$ ≈ T

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Let's Solve

Ex. Prove that relation R = $\{(x, y) : x$ is factor of $y\}$ is a transitive relation on integers.

Solution

Let $(a, b) \in R$ and $(b, c) \in R$

$\therefore b = ak_1$ and $c = bk_2$ for some $k_1, k_2 \in \mathbb{Z}$

$\Rightarrow c = ak_1k_2$

$\therefore a$ is a factor of c

$\therefore (a, c) \in R$

Hence it is transitive relation

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Quiz.



Quiz.

Chat Off

Q. Consider the following two binary relations on the set

$A = \{a, b, c\}$: $R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$

$R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$. Then:

A both R_1 and R_2 are not symmetric

B R_1 is not symmetric but it is transitive

C R_2 is symmetric but it is not transitive

D both R_1 and R_2 are transitive

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Quiz.

Chat On

Q. Consider the following two binary relations on the set $A = \{a, b, c\}$: $R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$ $R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$. Then:

A. both R_1 and R_2 are not symmetric

B. R_1 is not symmetric but it is transitive

C. R_2 is symmetric but it is not transitive

D. both R_1 and R_2 are transitive



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Quiz.

Chat On

Q. Consider the following two binary relations on the set $A = \{a, b, c\}$: $R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$ $R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$. Then:

Solution

$\because (b, c) \in R$, but $(c, b) \notin R_1$

$\rightarrow R_1$ is not symmetric

R_2 is symmetric but $(b, a) \in R_2$ and $(a, c) \in R_2$ but $(b, c) \notin R_2$

$\Rightarrow R_2$ is not transitive



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2 Types of Relations

(7) Equivalence Relation

Relation R defined on any set A is equivalence relation if

R is reflexive

R is symmetric

R is transitive

For example, let us Consider A = {1, 2, 3, 4, 5}
Defined as R = {(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (1, 5), (5, 1)} is equivalence relation.

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Quiz.



Quiz.

Chat Off

Q. The relation "less than" in the set of natural numbers is

A Only symmetric

B Only transitive

C Only reflexive

D Equivalence relation

(x, y) $x < y \Rightarrow y < x$

(x, x) $x < x \Rightarrow R$

$(x, y), (y, z) \Rightarrow (x, z)$

$x < y$ & $y < z$ $\Rightarrow x < z$



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Quiz.

Chat On

Q. The relation "less than" in the set of natural numbers is

A. Only symmetric

B. Only transitive

C. Only reflexive

D. Equivalence relation



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Quiz.

Chat On

Q. The relation "less than" in the set of natural numbers is

Solution

Since $x < y,\ y < z \Rightarrow x < z \forall\ x, y, z \in N$

$\therefore x\,R\,y, y\,R\,z \Rightarrow x\,R\,z, \therefore$ Relation is transitive,

$\therefore x < y$ does not give $y < x, \therefore$ Relation is not symmetric

Since $x < x$ does not hold, hence relation is not reflexive.



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Let's Solve



Let's Solve

Ex. A relation R is defined on set A of triangles in a plane as $R = \{(T_1, T_2) | T_1 \text{ is similar to } T_2\}$ Prove that it is equivalence relation.



$(T_1, T_2), (T_2, T_3) \Rightarrow (T_1, T_3)$

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Let's Solve

Ex. A relation R is defined on set A of triangles in a plane as $R = \{(T_1, T_2) | T_1 \text{ is similar to } T_2\}$ Prove that it is equivalence relation.

Solution

$(T_1, T_1) \in R \ \forall \ T_1 \in A$

Reflexivity : As T_1 is similar to T_1
So, R is reflexive relation

Let $T_1, T_2 \in A$
Such that $(T_1, T_2) \in R$

Symmetricity : $\Rightarrow T_1$ is similar to T_2

$\Rightarrow T_2$ is similar to T_1

$\Rightarrow (T_2, T_1) \in R$

So, R is Symmetric relation.

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Let's Solve

Ex. A relation R is defined on set A of triangles in a plane as $R = \{(T_1, T_2) | T_1 \text{ is similar to } T_2\}$ Prove that it is equivalence relation.

Solution

Let $T_1, T_2, T_3 \in A$

So that $(T_1, T_2) \in R$ and $(T_2, T_3) \in R$

$\Rightarrow T_1$ is similar to T_2 and T_2 is similar to T_3

Transitivity : $\Rightarrow T_1$ is similar to T_3 and $(T_1, T_3) \in R$

So, R is transitive relation.

So, R is an equivalence relation.

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Quiz.



Quiz.

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Q. Which of the following relations on A = {1, 2, 3} is reflexive?

A {(1, 2), (1, 3), (2, 1), (2, 2), (3, 3)}

B {(1, 1), (2, 3), (2, 1), (2, 2), (3, 3)}

C {(1, 1), (2, 3), (3, 1), (2, 2), (3, 2)}

D none of these



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Quiz.

Chat On

Q. Which of the following relations on A = {1, 2, 3} is reflexive?

A {(1, 2), (1, 3), (2, 1), (2, 2), (3, 3)}

B {(1, 1), (2, 3), (2, 1), (2, 2), (3, 3)}

C {(1, 1), (2, 3), (3, 1), (2, 2), (3, 2)}

D none of these



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Quiz.

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Q. Which of the following relations on A = {1, 2, 3} is reflexive?

Solution

R = {(1, 1), (2, 3), (2, 1), (2, 2), (3, 3)} is reflexive
as it contains all the elements (1, 1), (2, 2) and (3, 3)



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Quiz.

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Q. Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on A = {3, 6, 9, 12}. Relation R is _____.

A. Reflexive and transitive only

B. Reflexive only

C. Reflexive and symmetric only

D. An equivalence relation



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Quiz.

Chat On

Q. Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on A = {3, 6, 9, 12}. Relation R is _____.

A. Reflexive and transitive only

B. Reflexive only

C. Reflexive and symmetric only

D. An equivalence relation



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Quiz.

Chat On

Q. Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on A = {3, 6, 9, 12}. Relation R is _____.

Solution

Given A = {3, 6, 9, 12}
Since (3, 3), (6, 6), (9, 9), (12, 12) ∈ R
Thus its reflexive relation

Here, (6, 12) ∈ R, but (12, 6) ∉ R
Thus R is not symmetric

Here, (3, 6) ∈ R and (6, 12) ∈ R, also (3, 12) ∈ R
So relation is transitive also.



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